

Core Pure Mathematics 2

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Chapter 6: Hyperbolic Functions

Why this matters

Hyperbolic functions are a family of functions — $\sinh$, $\cosh$, $\tanh$ — that behave algebraically like trigonometric functions (they satisfy strikingly similar identities and calculus rules) but are defined using e^x rather than circles, and are connected to the hyperbola $x^2 - y^2 = 1$ in the same way $\sin/\cos$ are connected to the circle $x^2 + y^2 = 1$. They appear throughout further mechanics, further pure, and real-world modelling (e.g. the shape of a hanging chain, the catenary, is a $\cosh$ curve).

Concept: definitions of $\sinh$, $\cosh$, $\tanh$

$$\sinh x = (e^x - e^{-x})/2, \cosh x = (e^x + e^{-x})/2, \tanh x = \sinh x / \cosh x = (e^x - e^{-x})/(e^x + e^{-x})$$

Note the naming convention: "sinh" is read "shine" or "sinch", "cosh" rhymes with "gosh", "tanh" is "than" or "tanch" — these are separate functions, not $\sin/\cos$ multiplied by h .

Key vocabulary

Term	Meaning
$\sinh x$	$(e^x - e^{-x})/2$ — odd function, range is all real numbers
$\cosh x$	$(e^x + e^{-x})/2$ — even function, range is $[1, \infty)$
$\tanh x$	$\sinh x / \cosh x$ — odd function, range is $(-1, 1)$, horizontal asymptotes at $y = \pm 1$
arsinh , arcosh , artanh	Inverse hyperbolic functions ("area $\sinh$ " etc., named for a link to hyperbolic sector area)

Concept: the fundamental hyperbolic identity

Just as $\cos^2\theta + \sin^2\theta = 1$ for ordinary trig functions, hyperbolic functions satisfy an analogous identity — but with a crucial sign difference, because $\cosh$ and $\sinh$ parametrise a hyperbola ($x^2 - y^2 = 1$) rather than a circle:

$$\cosh^2 x - \sinh^2 x = 1$$

This single identity, together with the definitions, generates all the other identities you need (much like $\cos^2 + \sin^2 = 1$ does for ordinary trig).

Geometric picture. Just as $(\cos \theta, \sin \theta)$ traces the unit circle $x^2 + y^2 = 1$ with θ an angle, the point $(\cosh t, \sinh t)$ traces the right-hand branch of the unit hyperbola $x^2 - y^2 = 1$ as t ranges over all real numbers — that's immediate from $\cosh^2 t - \sinh^2 t = 1$. But t here is not an angle (a hyperbola has no natural angle to sweep). It turns out t equals *twice the area* of the hyperbolic "sector" bounded by the

curve, the x-axis, and the line from the origin to $(\cosh t, \sinh t)$ — the exact hyperbolic analogue of θ being twice the circular sector area for the unit circle. This is precisely why the inverse functions are called arsinh , arcosh , artanh ("area sinh" etc.): $\operatorname{arsinh} x$ returns the parameter t — twice that sector area — for which $\sinh t = x$.

Worked example 1 — Proving the fundamental identity from the exponential definitions

WORKED EXAMPLE 1

Question. Prove that $\cosh^2 x - \sinh^2 x = 1$ for all real x .

Step 1. Write $\cosh^2 x$ and $\sinh^2 x$ using the exponential definitions: **(M1)** $\cosh^2 x = [(e^x + e^{-x})/2]^2 = (e^{2x} + 2 + e^{-2x})/4$ $\sinh^2 x = [(e^x - e^{-x})/2]^2 = (e^{2x} - 2 + e^{-2x})/4$

Step 2. Subtract: **(M1)(A1)** $\cosh^2 x - \sinh^2 x = (e^{2x} + 2 + e^{-2x})/4 - (e^{2x} - 2 + e^{-2x})/4 = (2 - (-2))/4 = 4/4 = 1$

Answer: $\cosh^2 x - \sinh^2 x = 1$ ■ **(A1)**

EXAM TIP

If you are shaky on this, it is tempting to memorise identities in isolation. Instead, derive any hyperbolic identity you forget in an exam directly from the exponential definitions — it only takes 30 seconds and removes the risk of confusing a hyperbolic identity with its trig counterpart (getting a sign wrong, for instance).

Key formula — further hyperbolic identities (compare with the trig originals)

KEY FORMULA

Hyperbolic identity	Trig counterpart	Sign difference?
$\cosh^2 x - \sinh^2 x = 1$	$\cos^2 x + \sin^2 x = 1$	Yes
$\sinh 2x = 2 \sinh x \cosh x$	$\sin 2x = 2 \sin x \cos x$	No
$\cosh 2x = \cosh^2 x + \sinh^2 x = 2\cosh^2 x - 1 = 1 + 2\sinh^2 x$	$\cos 2x = \cos^2 x - \sin^2 x = 2\cos^2 x - 1 = 1 - 2\sin^2 x$	Yes (middle form)
$1 - \tanh^2 x = \operatorname{sech}^2 x$	$1 + \tan^2 x = \sec^2 x$	Yes

Concept: sketching the graphs of $\sinh$, $\cosh$, $\tanh$

- **$y = \sinh x$:** passes through the origin, odd function (symmetric under 180° rotation about the origin), increasing for all x , no asymptotes.
- **$y = \cosh x$:** passes through $(0, 1)$, even function (symmetric about the y-axis), minimum value 1 at $x = 0$, looks like a smooth "U" that grows like $\frac{1}{2}e^x$ for large x .
- **$y = \tanh x$:** passes through the origin, odd function, increasing, horizontal asymptotes $y = 1$ (as $x \rightarrow \infty$) and $y = -1$ (as $x \rightarrow -\infty$).

Concept: inverse hyperbolic functions and their logarithmic forms

Because $\sinh$, $\cosh$ (for $x \geq 0$), and $\tanh$ are one-to-one (on suitable domains), they have inverse functions, which can each be written as a natural logarithm — a genuinely useful and examinable result:

$$\begin{aligned}\operatorname{arsinh} x &= \ln(x + \sqrt{x^2 + 1}), \text{ for all real } x \\ \operatorname{arcosh} x &= \ln(x + \sqrt{x^2 - 1}), \text{ for } x \geq 1 \text{ (taking the positive branch)} \\ \operatorname{artanh} x &= \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right), \text{ for } -1 < x < 1\end{aligned}$$

Worked example 2 — Deriving the logarithmic form of $\operatorname{arsinh} x$

WORKED EXAMPLE 2

Question. Show that if $y = \operatorname{arsinh} x$, then $y = \ln(x + \sqrt{x^2 + 1})$.

Step 1. If $y = \operatorname{arsinh} x$, then $x = \sinh y = (e^y - e^{-y})/2$. **(M1)**

Step 2. Multiply both sides by $2e^y$ to clear the fraction: **(M1)** $2x e^y = e^{2y} - 1$, i.e. $e^{2y} - 2x e^y - 1 = 0$

Step 3. This is a quadratic in e^y . Let $u = e^y$: $u^2 - 2xu - 1 = 0$. Solve using the quadratic formula: **(M1)** $u = [2x \pm \sqrt{(4x^2 + 4)}]/2 = x \pm \sqrt{x^2 + 1}$

Step 4. Since $u = e^y > 0$ for all real y , and $\sqrt{x^2 + 1} > |x|$ always, we must take the $+$ sign (the $-$ sign would give a negative value): **(A1)** — **justifying the choice of root** $e^y = x + \sqrt{x^2 + 1}$

Step 5. Take natural logs of both sides: **(A1)** $y = \ln(x + \sqrt{x^2 + 1})$

Answer: $\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$ ■

EXAM TIP

The step most often botched is choosing the wrong root of the quadratic. Explicitly argue why e^y must be positive, and why only the "+" root guarantees that — this reasoning (not just quoting the final log form) is exactly what a "show that" question is testing, and skipping it loses marks even with the correct final answer.

Concept: calculus of hyperbolic functions

The derivatives and integrals of hyperbolic functions mirror the trig ones almost exactly, with the sign changes exactly matching the identity table above.

Key formula

KEY FORMULA

$f(x)$	$f'(x)$	$\int f(x) \, dx$
$\sinh x$	$\cosh x$	$\cosh x + c$
$\cosh x$	$\sinh x$	$\sinh x + c$
$\tanh x$	$\operatorname{sech}^2 x$	$\ln(\cosh x) + c$

$\operatorname{arsinh} x$	$1/\sqrt{x^2+1}$	—
$\operatorname{arcosh} x$	$1/\sqrt{x^2-1}$	—
$\operatorname{artanh} x$	$1/(1-x^2)$	—

Worked example 3 — Differentiating an inverse hyperbolic function

WORKED EXAMPLE 3

Question. Show that if $y = \operatorname{arcosh} x$ ($x > 1$), then $dy/dx = 1/\sqrt{x^2 - 1}$.

Step 1. If $y = \operatorname{arcosh} x$, then $x = \cosh y$. Differentiate implicitly with respect to x : (M1) $1 = \sinh y \cdot dy/dx$

Step 2. Rearrange: $dy/dx = 1/\sinh y$. (M1)

Step 3. Use $\cosh^2 y - \sinh^2 y = 1$ to write $\sinh y$ in terms of x : $\sinh^2 y = \cosh^2 y - 1 = x^2 - 1$, so $\sinh y = \sqrt{x^2 - 1}$ (taking the positive root since $y > 0$ for $x > 1$ means $\sinh y > 0$). (M1)(A1) — **correct use of the identity and correct choice of sign**

Answer: $dy/dx = 1/\sqrt{x^2 - 1}$ ■ (A1)

Worked example 4 — Integrating using a hyperbolic substitution

WORKED EXAMPLE 4

Question. Find $\int 1/\sqrt{x^2 + 4} dx$ using the substitution $x = 2 \sinh u$.

Step 1. Substitute $x = 2 \sinh u$, so $dx = 2 \cosh u du$, and $x^2 + 4 = 4\sinh^2 u + 4 = 4(\sinh^2 u + 1) = 4\cosh^2 u$ (using $\cosh^2 u - \sinh^2 u = 1$ rearranged). (M1)(M1)

Step 2. Substitute into the integral: (M1) $\int 1/\sqrt{4\cosh^2 u} \cdot 2\cosh u du = \int 1/(2\cosh u) \cdot 2\cosh u du = \int 1 du = u + c$

Step 3. Convert back to x using $u = \operatorname{arsinh}(x/2)$ (since $x = 2\sinh u \implies \sinh u = x/2 \implies u = \operatorname{arsinh}(x/2)$): (A1)

Answer: $\int 1/\sqrt{x^2 + 4} dx = \operatorname{arsinh}(x/2) + c$, equivalently $\ln(x/2 + \sqrt{x^2/4 + 1}) + c$.

EXAM TIP

This mirrors the trig substitution $x = a \sin \theta$ used for $\int 1/\sqrt{a^2 - x^2} dx$ from Core Pure — notice the parallel. The rule of thumb: $a^2 + x^2$ patterns suggest $\sinh$ substitutions (since $\cosh^2 - \sinh^2 = 1$ kills a plus sign), while $a^2 - x^2$ patterns still use ordinary $\sin/\cos$ substitutions. Recognising "which family of substitution matches which sign pattern" is the single most transferable skill in this section.

Chapter summary

Idea	Key point
Definitions	$\sinh x = (e^x - e^{-x})/2$, $\cosh x = (e^x + e^{-x})/2$, $\tanh x = \sinh x / \cosh x$
Fundamental identity	$\cosh^2 x - \sinh^2 x = 1$ (note the minus sign, unlike $\cos^2 + \sin^2 = 1$)
Inverse forms	$\operatorname{arsinh} x = \ln(x + \sqrt{x^2 + 1})$; $\operatorname{arcosh} x = \ln(x + \sqrt{x^2 - 1})$, $x \geq 1$; $\operatorname{artanh} x = \frac{1}{2} \ln((1+x)/(1-x))$, $ x < 1$
Derivatives	$d/dx(\sinh x) = \cosh x$; $d/dx(\cosh x) = \sinh x$; $d/dx(\tanh x) = \operatorname{sech}^2 x$
Substitutions	$x = a \sinh u$ for $a^2 + x^2$ integrals (mirrors $x = a \sin \theta$ for $a^2 - x^2$ integrals)

Common exam mistakes

- Confusing the sign in $\cosh^2 x - \sinh^2 x = 1$ with the trig version (writing a + instead of -).
- Forgetting that $d/dx(\cosh x) = \sinh x$, NOT $-\sinh x$ (unlike $d/dx(\cos x) = -\sin x$) — the hyperbolic derivatives do not carry the trig minus signs.
- Choosing the wrong root when deriving inverse hyperbolic log forms (forgetting to justify why e^y , or $\sinh y$, must be positive).
- Misremembering the domain restriction on $\operatorname{arcosh} x$ ($x \geq 1$) and $\operatorname{artanh} x$ ($-1 < x < 1$).
- Treating $\tanh x = \pm 1$ as achievable — $y = \pm 1$ are horizontal asymptotes only; $\tanh x$ gets arbitrarily close but never actually equals ± 1 , so $\operatorname{artanh}(\pm 1)$ is undefined.
- Sign errors when substituting $\cosh^2 u - \sinh^2 u = 1$ to simplify expressions like $\sinh^2 u + 1$ or $\cosh^2 u - 1$ mid-integration.

You should now be able to

- Define $\sinh$, $\cosh$, and $\tanh$ in terms of e^x , and sketch their graphs.
- Prove and use the identity $\cosh^2 x - \sinh^2 x = 1$ and related double-angle identities.
- Express the inverse hyperbolic functions as natural logarithms, and derive these forms from the exponential definitions.
- Differentiate and integrate hyperbolic and inverse hyperbolic functions, including using hyperbolic substitutions in integration.